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9.7 Comparison of Primary Structure – Dicot and Monocot Root, Stem and Leaf

Anatomy of Dicot and Monocot Roots

In different parts of the plants, the various tissues are distributed in characteristic patterns. This is best understood by studying their internal structure by cutting sections (transverse or longitudinal or both) of the part to be studied.

Primary Structure of Dicot Root – Bean Root

The transverse section of the dicot root (Bean) shows the following plan of arrangement of tissues from the periphery to the centre.

Piliferous Layer or Epiblema

The outermost layer of the root is called **piliferous layer or epiblema.** It is made up of single layer of parenchyma cells which are arranged compactly without intercellular spaces. It is devoid of epidermal pores and cuticle. It possesses root hairs which are single celled. It absorbs water and mineral salts from the soil. The chief function of piliferous layer is **protection.**

Cortex

Cortex consists of only parenchyma cells. These cells are loosely arranged with intercellular spaces to make gaseous exchange easier. These cells may store food reserves. The cells are oval or rounded in shape. Sometimes they are polygonal due to mutual pressure. Though chloroplasts are absent in the cortical cells, starch grain are stored in them. The cells also possess leucoplasts. The innermost layer of the cortex is endodermis. Endodermis is made up of single layer of barrel shaped parenchymatous cells. Stele is completely surrounded by endodermis. The radial and the inner tangential walls of endodermal cells are thickened with **suberin and lignin.** This thickening was first noted by **Robert Casparay** in 1965. So these thickenings are called **casparian strips.** But these casparian strips are absent in the endodermis cells which are located opposite the protoxylem elements. These thin-walled cells without casparian strips are called **passage cells** through which water and mineral salts are conducted from the cortex to the xylem elements. Water cannot pass through other endodermal cells due to the presence of casparian thickenings.

Check Your Grasp!

Give the exact location and function of passage cells?

In roots some cells of the endodermis usually the ones opposite to protoxylem, remain thin walled. These cells are called passage cells. They help in radial diffusion of water.

Stele

All the tissues present inside endodermis comprise the stele. It includes pericycle and vascular system.

Pericycle

Pericycle is generally a single layer of parenchymatous cells found inner to the endodermis. It is the outermost layer of the stele. Lateral roots originate from the pericycle. Thus, the lateral roots are endogenous in origin.